

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2457301.8342 +/- 0.0072 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.112 +/- 0.041
Transit depth (Rp/Rs)^2	1.26 +/- 0.91 [%]
Orbital Inclination (inc)	86.14 +/- 1.45
Airmass coefficient 1 (a1)	0.879 +/- 0.086
Airmass coefficient 2 (a2)	0.11 +/- 0.24
Scatter in the residuals of the lightcurve fit is	9.86 %
Best Comparison Star	#3 - [195, 312]
Optimal Aperture	1.82
Optimal Annulus	5.29
Transit Duration (day)	0.1454 +/- 0.0098

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.112 ± 0.041 vs 0.1036 (deviation 0.2σ)
Duration fit vs published	0.1454 d vs 0.1567 d (deviation 1.15σ)
Mid-time O-C	-12.6 min
Depth SNR	2.7
Residual scatter	9.86 %

Light curve

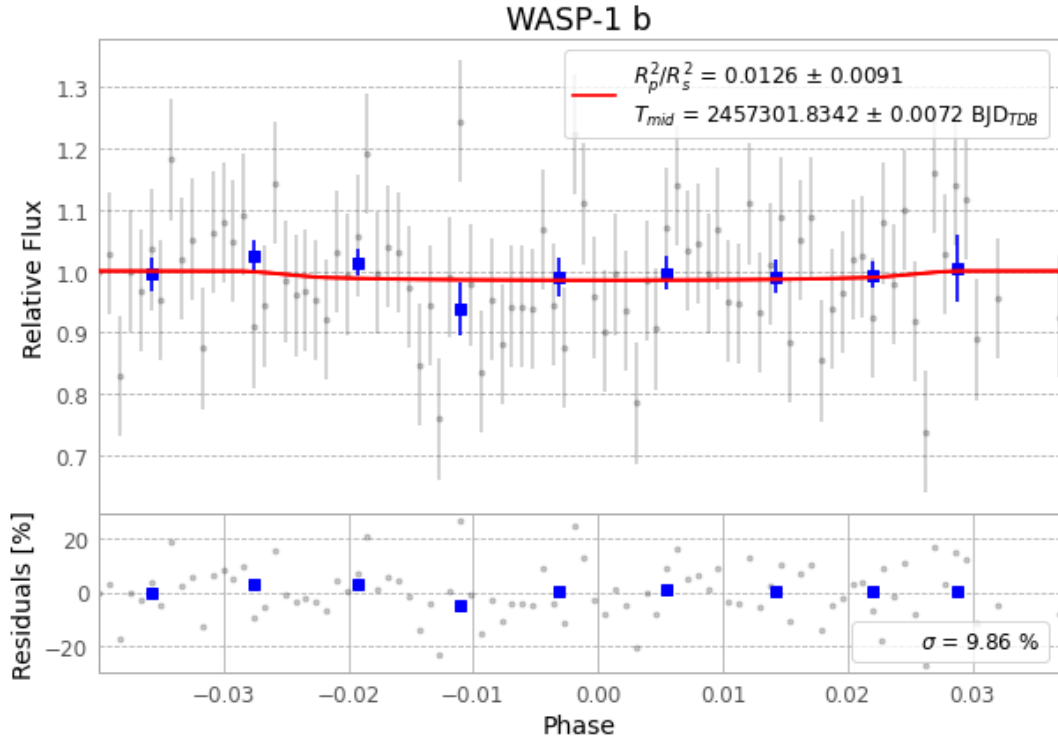


Figure 1 — FinalLightCurve_WASP-1 b_2015-10-05.png (SHA-256 25a2d223be1506f3...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [308, 258], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 a21816a1913f1b1c...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

107 FITS frames (70 MB), WASP-1151006052712.FITS → WASP-1151006104512.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-1-151006.log (4c255c1a241e...), FinalLightCurve_WASP-1 b_2015-10-05.png (25a2d223be15...), FinalLightCurve_WASP-1 b_2015-10-05.csv (1076cbe745c3...)

Compute resources

Wall time 185.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-01 03:20Z.